SIMATS ENGINEERING ASSESSMENT- TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4303

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Marks-90/100

COURSE OUTCOME COVERED

CO-1 : Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: **Total Marks: 30**

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly. Questions:

1. **Identify appropriate HTML5 semantic elements missing in the structure.**

Ans : Semantic elements such as <header>, <main>, and <section> are missing. They provide meaningful structure and improve accessibility.

2. **Modify the structure to make it standards-compliant.**

Ans : Use a valid HTML5 document structure with <!DOCTYPE html>, proper metadata, labels, and semantic elements. Close all tags correctly.

3. **Interpret the HTTP request generated during form submission.**

Ans : The POST method sends the form data to the /register server endpoint. The data is included in the request body rather than the URL.

4. **Analyze the HTTP response cycle from server to browser.**

Ans : The server receives and processes the request, then sends an HTTP response with a status code and content. The browser displays the resulting page or message.

5. **Suggest improvements for usability and accessibility.**

Ans : Add labels, required-field validation, clear instructions, and meaningful error messages. Use keyboard-accessible controls and appropriate input types.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body> </html>
```

Questions:

1. **Identify improper nesting and missing closing tags.**

The <h1> and <p> elements are missing their closing tags. The original HTML also has an incomplete structure that should be corrected.

2. **Analyze how different browsers may interpret this structure.**

Browsers may automatically correct malformed HTML differently, potentially causing inconsistent layouts. This can affect the display and document structure.

3. **Identify missing semantic elements (e.g., header, section).**

The original code lacks semantic elements such as <header>, <main>, and <section>. These elements improve page organization and accessibility.

4. **Provide a corrected W3C-compliant HTML structure.**

Use <!DOCTYPE html>, proper nesting, and matching opening and closing tags. Include valid HTML5 semantic elements and required metadata.

5. **Suggest techniques to ensure cross-browser compatibility.**

Use valid HTML and CSS, test the website in different browsers, and use CSS resets when needed. Follow web standards and use responsive design techniques.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Questions:

1. **Identify issues in the form configuration.**

The button uses type="button" instead of type="submit", so it does not submit the form automatically. The GET method is also unsuitable for transmitting passwords.

2. **Analyze why the HTTP request is not triggered.**

The type="button" control does not trigger form submission by default. Changing it to type="submit" allows the browser to submit the form.

3. **Interpret the role of GET method in this scenario.**

The GET method sends form data through URL query parameters. It is unsuitable for passwords because sensitive information may appear in browser history, URLs, and logs.

4. **Propose a corrected implementation.**

Use method="POST" and type="submit" in the form. Include named input fields and appropriate validation to submit the data correctly.

5. **Suggest improvements for secure data transmission.**

Use HTTPS to encrypt data during transmission and validate input on the server. Store passwords securely using a strong password-hashing algorithm.

Code of Conduct:

I _____Shanmukha.B(192372217)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shanmukha.B

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4– Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25% (22.5)	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25% (22.5)	Detailed, wellexplained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20% (18)	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15% (13.5)	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10% (9)	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5% (4.5)	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

1 Online Symposium Portal

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta  name="viewport"  content="width=device-width,  initial-scale=1.0">
   <title>Symposium Registration</title>
</head>
<body>
   <header>
      <h1>Online Symposium Portal</h1>
   </header>

   <main>
 <section>
    <h2>Student Registration</h2>
    <form  action="/register"  method="POST">
       <label for="name">Name:</label>
       <input type="text" id="name" name="name" required>

       <label for="email">Email:</label>
       <input type="email" id="email" name="email" required>

       <button type="submit">Submit</button>
    </form>
</section>
   </main>
</body>
</html>
```

Output :

Online Symposium Portal

Student Registration

Name: Email: Submit

2 Cross-Browser Compatibility Issue

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta  name="viewport"  content="width=device-width,  initial-scale=1.0">
    <title>Company Website</title>
</head>
<body>
    <header>
        <h1>Welcome</h1>
    </header>

    <main>
        <section>
            <h2>About Us</h2>
            <p>Welcome to our company website.</p>
        </section>
    </main>
</body>
</html>
```

Output :

Welcome

About Us

Welcome to our company website.

3 Form Submission Failure

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta   name="viewport"   content="width=device-width,   initial-scale=1.0">
    <title>Login Form</title>
</head>
<body>
    <header>
        <h1>User Login</h1>
    </header>

    <main>
        <section>
            <form action="/submit" method="POST">
                <label for="username">Username:</label>
                <input  type="text"  id="username"
```

```
            name="username" required>

        <label for="password">Password:</label>
        <input type="password" id="password"
            name="password" required>

        <button   type="submit">Login</button>
    </form>
        </section>
    </main>
</body>
</html>
```

Output :

